

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition? Answer Choices: A. It activates TLR5 on the surface of macrophages. B. It primarily induces the Th1-cell response. C. Peptidoglycan is its major antigen that induces an immune response. D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient presents with symptoms of community-acquired pneumonia and an X-ray showing patchy reticular opacities, which are consistent with bacterial pneumonia. The treatment with cephalexin, an antibiotic, suggests a bacterial etiology. Common bacterial pathogens causing community-acquired pneumonia include *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis*. These bacteria typically activate Toll-like receptors (TLR) on immune cells, particularly TLR4, which is important for their recognition. TLR5 is primarily involved in the recognition of flagellin in bacterial pathogens, which is not the case here. The response to bacteria is typically Th1-cell mediated, involving interferons and other cytokines. Peptidoglycan, which is a major component of the bacterial cell wall, is a key antigen in the immune response to these bacteria. Encapsulation, while used by some bacteria to evade the immune system (like in *Neisseria* species), is not the primary mechanism for the bacteria causing this type of pneumonia. </think> <answer> C. Peptidoglycan is its major antigen that induces an immune response. </answer>

